Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/22/2025
PI Name:	Konopka, Uwe Ph.D.		
Project Title:	Dusty Plasmas: Space Life and Physical Sciences and Research Apps		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Auburn	**State:**	AL
Zip Code:	36849-0001	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	EPSCoR-Physical Sciences NNH20ZHA001C
Start Date:	09/01/2020	**End Date:**	08/31/2022
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA EPSCoR
Contact Monitor:	Compton, Jeppie	**Contact Phone:**	
Contact Email:	jeppie.r.compton@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2022 (original end date was 8/31/2021) per NSSC information (Ed., 9/1/21)		
Key Personnel Changes/Previous PI:	Dr. Lawrence Dale Thomas serves as the Alabama NASA EPSCoR Director. Dr. Thomas is the administrative Principal Investigator (PI) on this proposal and will oversee the overall progress of the work and report to NASA.		
COI Name (Institution):	Hahn, Inseob Ph.D. (NASA Jet Propulsion Laboratory)		
Grant/Contract No.:	80NSSC20M0131		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Established Program to Stimulate Competitive Research (EPSCoR) Project Dusty plasmas are four-component plasma systems consisting of the standard plasma constituents of electrons, ions, and neutral atoms, with the addition of a fourth component: charged, solid, nanometer-to-micrometer-sized particulates (i.e., the "dust" particles). In both laboratory and space environments, the dust particles become charged through the acquisition of electrons and ions from the background plasma as well as various ionizing processes such as thermionic emission or photoelectron emission. Regardless of the charging processes, the dust particles are coupled to and become part of the plasma through their charge. However, the small charge-to-mass ratio of the dust grains (relative to the electrons or ions) means that the plasma process of the dust component is slowed to time (~10-2 s) and space (~10-3 m) scales that enable simultaneous studies of plasma dynamics in both the kinetic (particle) and fluid (collective) regimes. Moreover, the small charge-to-mass ratio also allows the thermal state of the dust component to be experimentally "tuned" over a range of Coulomb-coupling parameters (i.e., Gamma (G), ratio of electrostatic-to-thermal energy)) from the weakly-coupled (G << 1) to strongly-coupled (G >>1) regimes. This provides opportunities to explore regimes of plasma behavior that are generally difficult to achieve in standard plasmas. This proposed project seeks to perform a new investigation of controlled dusty plasma particle transport from weakly-coupled to strongly-coupled regimes. This work leverages the extensive expertise and experimental capabilities of the Auburn Dusty Plasma research group in laboratory and microgravity studies of dusty plasmas. In this work, two main activities are proposed for this one-year activity: (a) demonstration of "passively-driven, long-range" dust particle transport (i.e., transport distances >> collision mean free path) using biased electrodes to establish cyclical particle motion between several suspended dust clouds. (b) development of a programmable, segmented electrode with the ability to provide steady-state or periodic control of particle transport over long distances in a plasma. The long-term goals of this work are to develop a comprehensive understanding of dust particle transport over a broad range of experimental conditions that can lead to scaled studies of charged dust transport in environments relevant to lunar and Martian conditions. From these studies, it will be possible to provide a scientific basis for the next generation of remote and manned missions on airless and plasma exposed bodies throughout the solar system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	See Bibliography for publications/presentations resulting from this EPSCoR project. (Ed., 7/22/25)
Bibliography Type:	Description: (Last Updated: 09/30/2025)
Abstracts for Journals and Proceedings	Nofs L, Konopka U, Loch S. "Dust transport by induced plasma modulations in low pressure RF argon plasma." 2024 Dusty Plasma Workshop, Minneapolis, Minnesota, May 19-22, 2024. Abstracts. 2024 Dusty Plasma Workshop, Minneapolis, Minnesota, May 19-22, 2024. , Jan-2025
Abstracts for Journals and Proceedings	Nofs, L., Konopka, U., Loch, S.D. "Dust transport in a low temperature plasma device by induced plasma modulations." 65th Annual Meeting of the APS Division of Plasma Physics, Denver, Colorado, October 30-November 3, 2023. Abstracts. 65th Annual Meeting of the APS Division of Plasma Physics, Denver, Colorado, October 30-November 3, 2023. Session BP11.00037. https://meetings.aps.org/Meeting/DPP23/Session/BP11.37 , Oct-2023